

PH 1023 – GENERAL PHYSICS LABORATORY II

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THE 1 - Investigating the Viscosity of Coconut Oil via Air Bubble Kinetics

Viscosity is a measure of a liquid's resistance to flow. It describes how "thick" or "thin" a liquid is, essentially capturing the internal friction within the fluid. In liquids with high viscosity, like honey, molecules have more friction and move past each other more slowly, resulting in a slower flow. Conversely, liquids with low viscosity, like water, have less internal resistance, allowing them to flow more easily.

The viscosity of a liquid depends on several factors, including temperature, where an increase in temperature generally decreases viscosity because molecules move more freely. Viscosity plays a critical role in various applications, from lubrication in machinery to controlling fluid flow in industrial processes and even in biological systems, like blood flow. It's a key property in fields like materials science, engineering, and chemistry, providing insights into the behavior of fluids under different conditions.

Viscosity can be thought of as an index of quality to control applications using oils, paints and other fluids where the liquid flow is considered as a crucial property, which makes measuring viscosity vital. There are several methods to measure viscosity even in the commercial aspect such as capillary viscometer, vibrational viscometer, rotational viscometer and falling sphere viscometer. In this experiment a novel method is experimented to determine the viscosity of a liquid, using a rising air bubble which can be considered as an extension or a development of the falling sphere method.

Viscosity, η can be defined as,

$$\eta = \frac{F/A}{dv_x/dz}$$

Where (F/A) is the shearing stress and (dv_x/dz) is the velocity gradient in a fluid.

Consider a floating air bubble (a sphere of radius a) traveling upwards through a liquid at velocity (v) where viscosity is dominant, and turbulence can be neglected.

Preparatory Questions

- 1 Write an equation to the viscous drag force acting opposite to the direction of motion of the air bubble which travels upward direction of a liquid. (HINT: Use stokes law)
- 2 Hence derive an equation for the terminal velocity of the air bubble.
- 3 What are the assumptions that you made while deriving the equations.

Procedure

Apparatus - A long transparent bottle, a light source, a Smartphone, Viscous liquid(Coconut oil), Rubber tube

Steps -

1. Take a long transparent bottle and a rubber tube.
2. Make a hole in the lid of the bottle and insert one end of the rubber tube into the bottle.
3. Make a small hole in the bottom of the bottle to maintain the atmospheric pressure inside the bottle once it turns upside down.
4. Then fill the bottle with the coconut oil and quickly turn the bottle upside down.
(You should find a method to hang the bottle vertically)

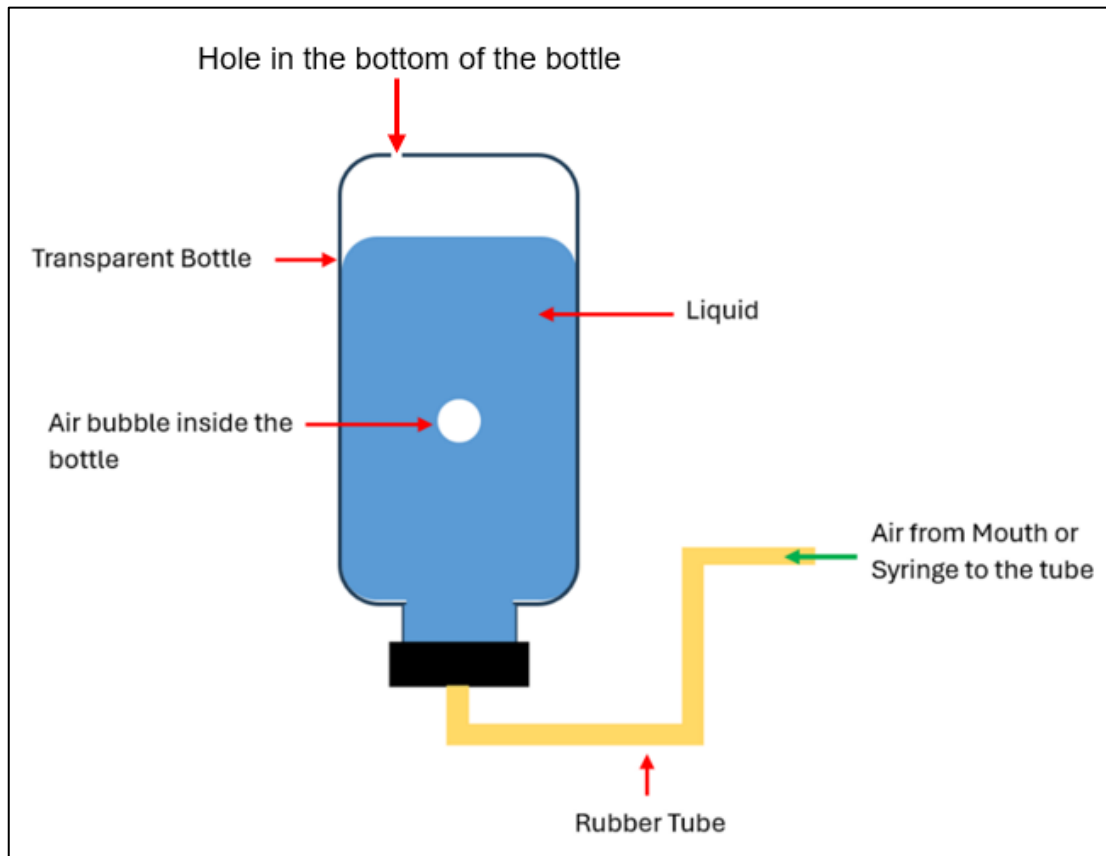


Figure 1 : Schematic of the Experimental Setup

5. Attach a measuring tape to the bottle as a reference measuring tool.
6. Mount the camera (smartphone with video facility) so that the camera doesn't shake or move.
7. Keep the bottle filled with the viscous liquid in front of the camera, so that the camera has a fully focused view of the bottle. Keep the light source behind the bottle.

Note - Record the video with the best video settings you have so that air bubbles are well focused, and the video is well stabilized.

8. After arranging the above setup, make the bubbles inside the liquid and record the motion of the bubbles. Among the bubbles, only one or few of them may attain the terminal velocity. You can connect an injection syringe to the other end of the rubber tube or use your mouth to make air bubbles inside the coconut oil.

9. Then insert the video into Tracker software and use it to obtain the velocities of the air bubbles.

Note - When choosing air bubbles with different radius, choose them so that the spread is maximized.

10. Measure the radius of the air bubbles that were chosen, with the terminal velocity and fill the table 1.1.

Note - To take the radius of the air bubbles go to “create -> measuring tools -> circle fitter”

Table 1.1 Bubble radius(a) and Terminal velocity (v)

Air bubble Number	Radius (a)	δa	V_{terminal}	δv
1				
2				
3				
..				
..				

11. Extract information from the above table and appropriately identify, modify and plot dependent and independent variables in order to measure the coefficient of viscosity.

12. Compare the measured coefficient of viscosity with a standard value.